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THE FILAMENTOUS GREEN ALGA *CLADOPHORA GLOMERATA* AS A HABITAT FOR LITTORAL MACROFAUNA IN THE NORTHERN BALTIC SEA

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ABSTRACT

Opportunistic filamentous algae are known to increase due to elevated nutrient levels (eutrophication) and to outcompete slow-growing perennial macroalgae in shallow coastal areas causing severe shifts in algal communities and habitat changes for the associated fauna. When the algae detach from the primary substrate and form sublittoral drifting mats, oxygen deficiency occurs frequently. We investigated the temporal succession of macrofauna in attached and loose-lying *Cladophora glomerata* to establish the role of this transient substrate. We also conducted a series of aquarium experiments using clean sea-water and water from fresh and decaying *C. glomerata* to study preferences among juvenile and adult crustaceans (*Gammarus* spp. and *Idotea baltica*). Both attached and loose-lying littoral *Cladophora glomerata* hosted rich macrofauna assemblages with significantly higher faunal abundance and biomass among the attached algae and clear temporal trends. The dominant taxa were the gastropods *Hydrobia* spp., the blue mussel *Mytilus edulis*, the crustaceans *Gammarus* spp. and *Idotea baltica*, as well as chironomids and oligochaetes. Aquarium experiments demonstrated that juvenile and adult *I. baltica* and juvenile *Gammarus* spp. preferred clean water and water from fresh algae to water from decomposing algae, whereas adult *Gammarus* spp. were indifferent. Our results demonstrate that both attached and drifting *C. glomerata* serve as useful habitats for littoral macrofauna.

Key Words: Drifting algae, Epifauna, *Gammarus*, *Idotea*, Rocky shores, Coastal eutrophication, Marine biodiversity

INTRODUCTION

In summer, the filamentous green alga *Cladophora glomerata* (L.) Kütz. and the canopy-forming brown alga *Fucus vesiculosus* L. dominate at shallow sites in the semi-enclosed non-tidal low-saline Baltic Sea (Wallentinus, 1991). Together they build a mosaic of microhabitats on littoral rocky shores and have been classified as the most productive subsystems of the Baltic Sea with algal production often exceeding 500 g C m^{-2} and y^{-1} (Guterstam *et al.*, 1978; Wallentinus, 1991). These algae are utilised by many small benthic animals, such as crustaceans, gastropods, cockles and mussels (Jansson, 1967; Fagerholm, 1975; Haage, 1975), which are essential links in the food webs of littoral ecosystems.

The escalating eutrophication alters the dominance patterns among marine algae, which affects associated fauna (Kotta *et al.*, 2000). Fast-growing annual algae like *Cladophora glomerata* increase and outcompete slow-growing perennial algae such as *Fucus vesiculosus* (Wallentinus, 1984; Fletcher, 1996). The increased algal production also causes increased export of drift algae, with *C. glomerata* being one of the most abundant species (Vahteri *et al.*, 2000; Norkko *et al.*, 2000; Berglund *et al.*, 2003). Drifting filamentous algae have been described as a serious threat to benthic communities causing hypoxia and anoxia and further hydrogen sulphide production (Norkko & Bonsdorff, 1996; Raffaelli *et al.*, 1998; Karlson *et al.* 2002) and are often perceived as a nuisance (Lehvo & Bäck, 2001). Examples of positive interactions are that drift algae serve as alternative habitats for sediment fauna, as food resources and as "rafts" for animal dispersal (Norkko *et al.*, 2000; Brooks & Bell, 2001; Salovius & Bonsdorff, 2004).

A change in the composition of benthic algal species affects macrofauna due to the altered quality of food (type of species and nutrient content) and changes in habitat (Seed & O'Connor, 1981; Kraufvelin *et al.*, 2002). Mu-

tualistic relationships exist between littoral fauna and macroalgae (see reviews by Jernakoff *et al.* 1996 and Arrontes 1999). Parallel to the consumption of algal matter by grazers, which may damage and even kill the algae, the algae trap particulate organic matter, which is also consumed by invertebrates (Edgar & Klumpp, 2003). The algae offer shelter from wave action (Salemaa, 1986), desiccation, and predation (Stoner, 1985). Conversely, the fauna mineralise nitrogen and phosphorous from organic matter, which favours algal growth (Jansson, 1974). The animals also clean macroalgal filaments of diatoms and epiphytic macroalgae, which increases algal photosynthesis (Brawley & Adey, 1981; Karez *et al.*, 2000). Grazing may further enhance algal spore release and dispersal (Buschmann & Bravo, 1990).

Eutrophication related research has been intense in the Baltic Sea since the 1970s and many harmful effects of nutrient enrichment have been listed. Over 1170 references concerning nutrient overload are found from the Baltic area, ranging from Kattegat in the southwest to the Bothnian Bay in the north (Rönnberg, 2001). Of these studies, 24 are concerned with eutrophication and macroalgae and even fewer with filamentous algae. Only Jansson (1967) and Fagerholm (1975) have provided quantitative data on macrofauna assemblages in attached filamentous algae from the Baltic Sea littoral zone (literature search in ISI Web of Science). Further, little information is available on macrofauna associated with filamentous algae from fully marine areas (Seed & O´Connor, 1981; Edgar, 1983; Preston & Moore, 1988; Gee & Warwick, 1994).

Coastal ecosystem deterioration is one of the most serious threats for the marine environment (Gray *et al.*, 2002) and functional shifts caused by structural changes are to be expected. Coastal eutrophication (Cloern, 2001) increases growth of annual filamentous algae, which retard perennial species, known to be very important habitats for macrofauna. During this change in algal communities the fauna diversity decreases (Kotta *et al.*, 2000). We analysed the spatial and temporal variation in faunal assemblages associated with attached and loose-lying *Cladophora glomerata* to establish the role of this transient substrate. Furthermore we conducted preference studies to test if juvenile and adult crustaceans chose actively between seawater, water from sessile assimilating algae and water from drifting decaying algae.

RESULTS

Attached *Cladophora glomerata* in the field. The mean biomass of *Cladophora glomerata* per m^2, calculated from individual samples, differed significantly between the four locations ($p = 0.005$ in a 2-way factorial ANOVA, df = 3, 9), but not over time ($p = 0.714$, df = 3, 9). Neither was there any significant interaction between the two random factors Location and Time ($p = 0.783$, df = 9, 32). Mean biomass (± SE, n = 3 individual replicates) for *C. glomerata* ranged between 20.7 ± 5.4 and 131.4 ± 71.3 g dwt m^{-2} (Fig. 1). The grand mean (± SE, n = 48 algal samples) for attached *C. glomerata* was 68 ± 7 g dwt m^{-2}.

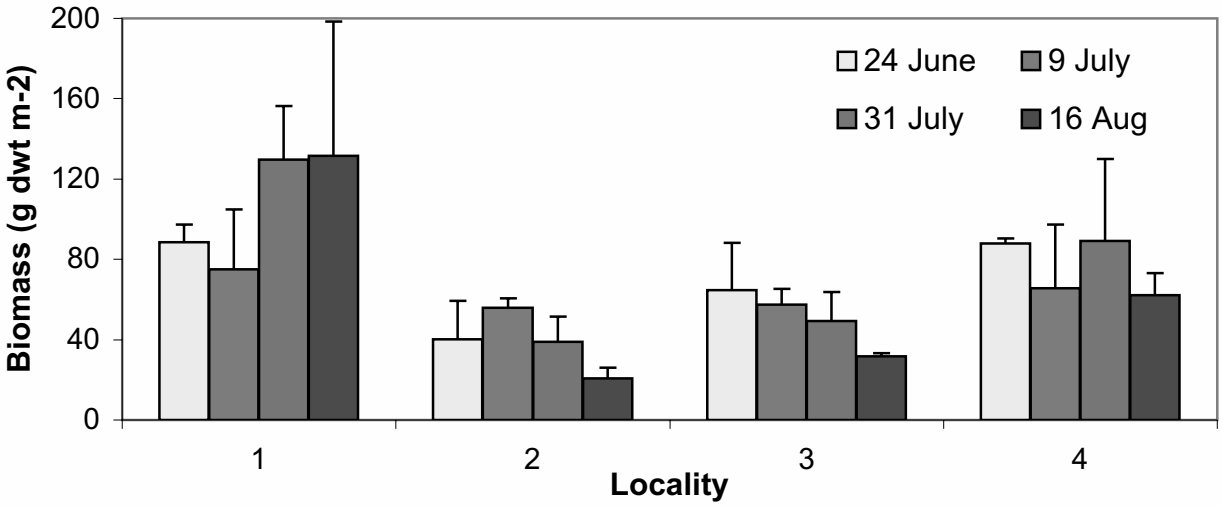

Fig.1. *Cladophora glomerata*. Temporal succession in biomass (g dwt m^{-2} ± SE) at four field localities.

Table 1. Mean abundance per 10 g dwt algae of invertebrate taxa present at the four sampling dates in attached and drifting *C. glomerata*.

Taxon		24 June		9 July		31 July		16 Aug	
		attached	drifting	attached	drifting	attached	drifting	attached	drifting
Hydrozoa									
	Hydridae	0	0	0	0	0	12	9	3
Nemertinea									
	Prostoma obscurum Schulze	2	0	0	0	0	0	0	0
Nematoda		82	407	0	0	0	0	0	0
Turbellaria									
	Planaria spp.	24	6	11	1	5	0	14	0
Arachnida									
	Acarina spp.	12	54	0	0	0	2	0	0
Polychaeta									
	Nereis diversicolor (O.F. Muller)	1	0	0	2	0	1	1	1
Oligochaeta spp.		210	928	83	52	5	20	0	3
Mollusca									
	Theodoxus fluviatilis (L.)	70	34	148	1	91	8	67	19
	Lymnaea peregra (Möller)	7	0	38	14	22	39	57	21
	Lymnaea stagnalis (L.)	0	0	3	0	0	1	5	1
	Hydrobia spp.	415	79	341	9	1568	140	1436	101
	Limapontia capitata (Müller)	0	0	0	0	0	0	56	6
	Bithynia tentaculata (L.)	2	5	0	0	0	0	0	0
	Macoma balthica (L.)	6	0	6	0	0	0	0	0
	Mytilus edulis L.	120	39	170	12	224	61	251	86
	Cerastoderma glaucum (Poiret)	6	0	36	22	638	47	179	38
Crustacea									
	Palaemon adspersus (Rathke)	0	0	0	0	0	0	9	7
	Mysis relicta Lovén	0	0	4	0	4	11	0	1
	Gammarus spp.	180	627	280	200	285	129	155	296
	Jaera albifrons Leach	83	8	208	3	110	3	28	0
	Idotea baltica Pallas	11	5	216	62	246	54	136	12
	Argulus foliaceus (L.)	0	0	2	0	0	0	0	0
	Copepoda spp.	50	320	46	47	0	43	0	0
	Daphnia sp.	2	0	0	0	0	0	0	0
	Balanus improvisus Darwin	6	0	8	0	12	1	6	1
Bryozoa									
	Electra crustulenta (Pallas)	+	+	+	+	+	+	+	+
Insecta spp.									
	Chironomus spp.	956	1979	657	656	607	542	337	559
	Cercopagis pengoi (Ostroumov)	0	0	0	0	0	0	309	240
	Palpomyia sp.	0	13	6	9	8	40	0	1
	Platambus maculatus (L.)	1	0	0	0	0	0	0	0
	Mycetobia pallipes Meigen	1	0	0	0	0	0	0	0
	Deronectes depressus (Fab.)	2	0	5	0	6	0	0	0
	Dytiscus caesus Duftschmid	2	0	0	0	0	0	0	0
	Tabanus sp.	0	1	1	1	0	0	0	4
	Agabus uliginosus (L.)	0	0	0	6	0	2	0	0
	Hydrobius fuscipes (L.)	0	3	0	3	0	0	0	0
	Agraylea multipunctata Curtis	0	0	13	5	9	8	32	1
	Insecta indetermined	2	5	6	75	2	4	7	0
Pisces									
	Gasterosteus aculeatus L.	12	0	1	3	0	0	0	0
	Pungitius pungitius (L.)	0	0	1	0	2	0	0	0
	Pomatoschistus minutus (Pallas)	0	0	0	0	9	1	5	0
	Number of taxa	27	18	25	21	20	23	21	21

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Table 2. Results of a 3-factor mixed model ANOVA on differences in animal a) community and b) population variables between attached and drifting *Cladophora glomerata* at four localities over time. All variables except Shannon-Wiener diversity and *Gammarus* spp. were transformed by ln(x + 1). L = Locality, A = Algae, T = Time, * means that $0.01 < p \leq 0.05$, ** means that $0.001 < p \leq 0.01$, *** means that $p \leq 0.001$. Values significant at the 0.05-level after correction for the number of tests by Hochberg's sequential Bonferroni are underlined.

a) Community variables

		Total abundance		Total biomass		Shannon diversity	
Source	df	MS	F	MS	F	MS	F
L	3	2.30	2.78	2.59	**20.22*** **	0.97	3.02
A	1	3.88	No test	24.92	No test	7.99	No test
T	3	3.27	**3.96***	1.11	**8.65** **	0.58	1.79
L×A	3	2.30	2.21	0.65	1.37	0.31	1.57
L×T	9	0.83	1.65	0.13	0.26	0.32	**2.05***
A×T	3	2.81	2.71	0.72	1.52	0.31	1.58
L×A×T	9	1.04	**2.08***	0.48	0.97	0.20	1.28
Residual	64	0.50	0.49	0.16			
Total	95						

b) Population variables

		Gammarus abundance		*Idotea* abundance		*Hydrobia* abundance		*Mytilus* abundance	
Source	df	MS	F	MS	F	MS	F	MS	F
L	3	2.85×10^5	2.53	12.34	2.34	4.66	0.96	27.84	**14.89*** **
A	1	1.86×10^5	No test	52.60	No test	196.30	No test	25.93	No test
T	3	1.97×10^5	1.75	41.80	**7.94** **	7.52	1.55	9.46	**5.06***
L×A	3	4.70×10^4	0.51	2.33	1.23	3.71	0.75	6.38	0.78
L×T	9	1.13×10^5	1.15	5.26	**2.63***	4.86	1.77	1.87	0.77
A×T	3	4.38×10^5	**4.75***	3.37	1.78	9.65	1.96	8.98	1.09
L×A×T	9	9.22×10^4	0.94	1.90	0.95	4.93	1.79	8.21	**3.40***
Residual	64	9.82×10^4	2.00	2.75	2.42				
Total	95								

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Macrofaunal community structure of attached and drifting *Cladophora glomerata*. Both attached and drifting *Cladophora glomerata* hosted rich macrofauna communities throughout the study period. The mean faunal abundance (± SE, n = 48 samples) were 2870 ± 460, and the mean biomass 12.7 ± 2.5 g wwt per 10 g dwt of attached algae. For drifting algae, the corresponding numbers were 2060 ± 280 individuals and 2.9 ± 0.5 g wwt per 10 g dwt of algae. The most numerous taxa overall were the gastropods *Hydrobia* spp., *Theodoxus fluviatilis* (L.) and *Lymnaea* spp., the bivalves *Mytilus edulis* L. and *Cerastoderma glaucum* (Poiret), the crustaceans *Gammarus* spp., *Idotea baltica* and *Jaera albifrons* Leach, as well as chironomids and oligochaetes. The full species list comprised an additional 30 taxa (Table 1). The molluscs contributed more to the total biomass than the crustaceans, which mainly consisted of juveniles. Although sometimes present in high numbers, neither chironomids nor oligochaetes contributed much to the total faunal biomass.

Total abundance, biomass and Shannon-Wiener diversity (log-base 2) had higher values in attached algae compared to drifting algae, although it could not be tested for Algae as a main effect in the 3-factor mixed model ANOVA (Table 2a, Fig. 2). There was, however, a significant interaction Algae x Location x Time for total abundance, although this effect disappeared after correcting for the numbers of comparisons. For Shannon-Wiener diversity there was a significant interaction between Location and Time, whereas for total biomass both main effects Location and Time were significant. The total biomass p-values also sustained the Bonferroni correction ($p < 0.001$ for Location and $p < 0.05$ for Time).

The abundance and biomass of *Idotea baltica*, *Hydrobia* spp. and *Mytilus edulis* were also higher in attached algae (biomass more evidently than abundance). *Gammarus*, on the other hand, seemed to be indifferent with regard to the algae (Table 2b, Fig. 2 for abundance data), with the exception of the June sampling, when there were significantly more *Gammarus* in

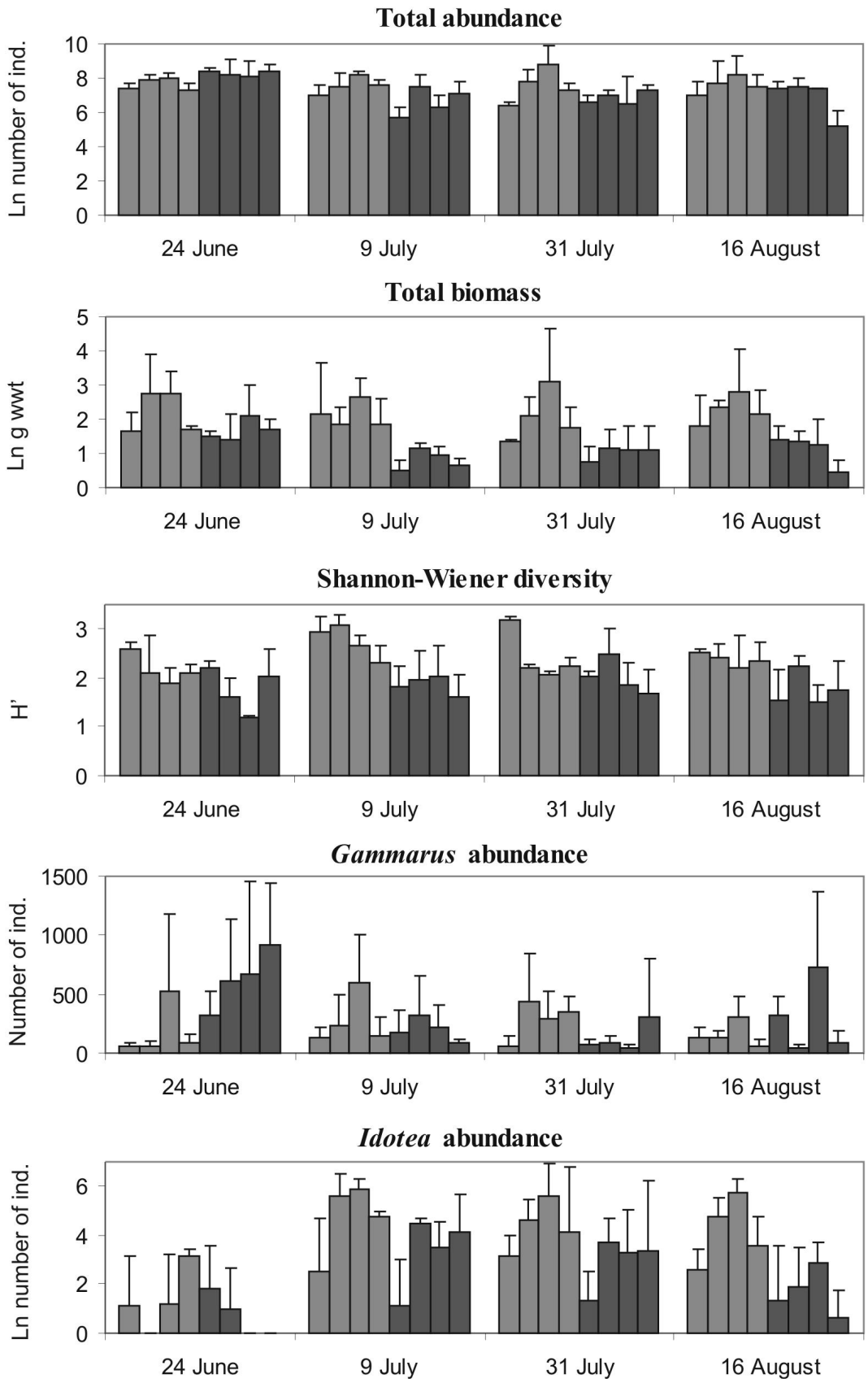

Fig. 2. *Cladophora glomerata*. Temporal differences in faunal community and population variables between attached (light bars) and drifting algae (dark bars), the four bars of each type at each occasion refer to the four sampling sites. Untransformed mean values (Shannon diversity and *Gammarus*) and values transformed by ln (x+1) (total abundance, total biomass and *Idotea*) ± SE are shown.

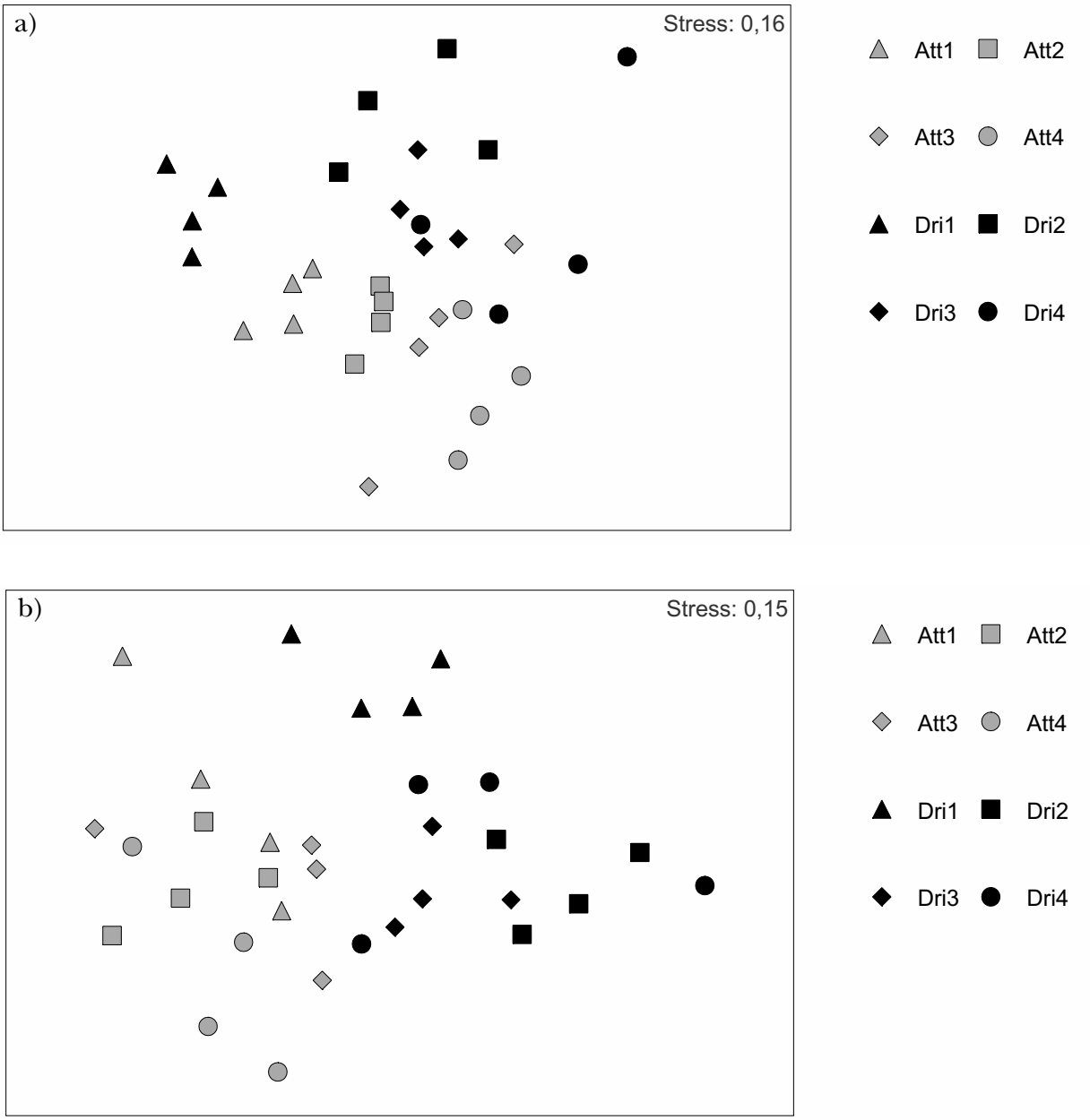

Fig. 3. NMDS ordination of square-root transformed animal a) abundance (stress 0.16) and b) biomass (stress 0.15) data. The attached algae (Att1-Att4) are represented by light symbols, whereas the drifting algae (Dri1-Dri4) are represented by dark symbols. The sampling occasions were 1 = 24th June, 2 = 9th July, 3 = 31st July and 4 = 16th August. Each symbol represents the mean of three replicates.

drifting algae than in attached algae (SNK-test after significant Algae × Time effect). The 3-factor mixed model ANOVA presented a significant Location x Time interaction for *I. baltica* as well as a significant Location x Algae x Time interaction for *M. edulis* abundance, whereas no significant results occurred for *Hydrobia*. The mean square for Algae as a main effect was more than ten times higher than all other mean squares for *Hydrobia*, but due to the study design no tests were available. For *I. baltica*, SNK-tests revealed that the observed differences were mainly due to lower abundance in samples from June and from fewer specimens at one of the four sites. For *M. edulis*, SNK-tests revealed that there were significantly higher abundances in attached algae than in drifting algae at all locations, in addition to significant differences between locations and sampling occasions.

Table 3. Each species' contribution (δ_i) to the average dissimilarities between attached (n=48) and drifting (n=48) algae for square-root transformed a) abundance (total δ% =58.41) and b) biomass data (total δ% = 63.46%).

a) Abundance

Taxonomic group	Average abundance attached	Average abundance drifting	Average term δ_i	Ratio Diss. / SD	Percent	Cumulative percent
***Hydrobia* spp.**	**939.87**	82.22	7.95	1.28	13.60	13.60
Chironomidae	639.23	**934.02**	7.18	1.24	12.29	25.89
***Gammarus* spp.**	224.95	**312.88**	4.50	1.12	7.71	33.60
Oligochaeta	74.64	**250.81**	4.07	0.90	6.97	40.58
Mytilus edulis	**191.08**	49.77	3.64	1.26	6.22	46.80
Idotea baltica	**151.97**	33.34	3.50	1.11	5.99	52.79
Jaera albifrons	**107.24**	3.40	3.36	1.13	5.75	58.53
Theodoxus fluviatilis	**94.13**	15.61	3.27	1.43	5.59	64.12
Cerastoderma glaucum	**214.84**	26.72	3.24	0.96	5.55	69.67
Lymnaea peregra	**33.29**	18.93	1.92	1.15	3.29	72.96

b) Biomass

Taxonomic group	Average biomass attached	Average biomass drifting	Average term δ_i	Ratio Diss. / SD	Percent	Cumulative percent
***Hydrobia* spp.**	**3.96**	0.35	10.97	1.36	17.29	17.29
Mytilus edulis	**3.38**	0.42	7.57	0.81	11.94	29.23
Theodoxus fluviatilis	**1.06**	0.18	7.40	1.29	11.67	40.90
Idotea baltica	**0.71**	0.16	5.53	1.24	8.71	49.60
***Gammarus* spp.**	0.73	**0.96**	5.46	1.12	8.60	58.21
Lymnaea peregra	**0.98**	0.09	3.54	0.72	7.65	65.86
Jaera albifrons	**0.26**	0.01	3.49	1.14	5.49	71.35
Chironomidae	0.32	**0.46**	3.46	1.22	5.46	76.81
Cerastoderma glaucum	**0.33**	0.06	3.07	0.97	4.84	81.65
Balanus improvisus	**0.19**	0.01	2.18	0.59	3.43	85.08

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Two special features deserve to be highlighted in this data (Fig. 2). First, the results for *Idotea baltica* were consistent over time, i.e. as juveniles appeared in the material (starting in July), the differences between algal habitat and localities were constant. Second, *Gammarus* (mostly > 50 individuals per 10 g dwt algae) exhibited no consistent patterns over time, nor any clear preferences for any of the investigated factors, with the exception of significantly more individuals in drifting algae compared to attached algae in June.

When examined by non-parametric multivariate techniques, the differences in macrofauna abundance and biomass between the two algal habitats (attached versus drifting), as well as among sampling occasions, became more evident. The analyses revealed seasonal patterns in community succession. Global tests of differences between attached and drifting algae (2-way crossed ANOSIM) gave p-values below 0.001 for both abundance and biomass data. Also for the temporal differences the global p-values were below 0.001. Pair-wise tests showed that sampling occasions more distant in time generally had a greater difference in community structure. These differences are visualised by NMDS-ordinations of both abundance and biomass. For the abundance data (Fig. 3a), the grouping is distinct for both the factors Algae and Time. Attached algae may be found in the lower parts and drifting algae in the upper parts of the figure, while samples from June cluster out to the left, July samples in the middle and samples from August to the right. From the abundance figure, it is also evident that the differences between times (especially differences between June-samples and other samples) tend to be bigger than differences between attached and drifting algae. For biomass data (Fig. 3b), the differences between attached and drifting algae were more evident than for abundance data, while the temporal trend was unclear.

Species contributing most to the dissimilarities between attached and drifting *Cladophora glomerata* are listed separately for abundance (dissimilarity = 58.41%) and biomass analyses (dissimilarity = 63.46%) (Table 3). The general abundance pattern revealed higher numbers

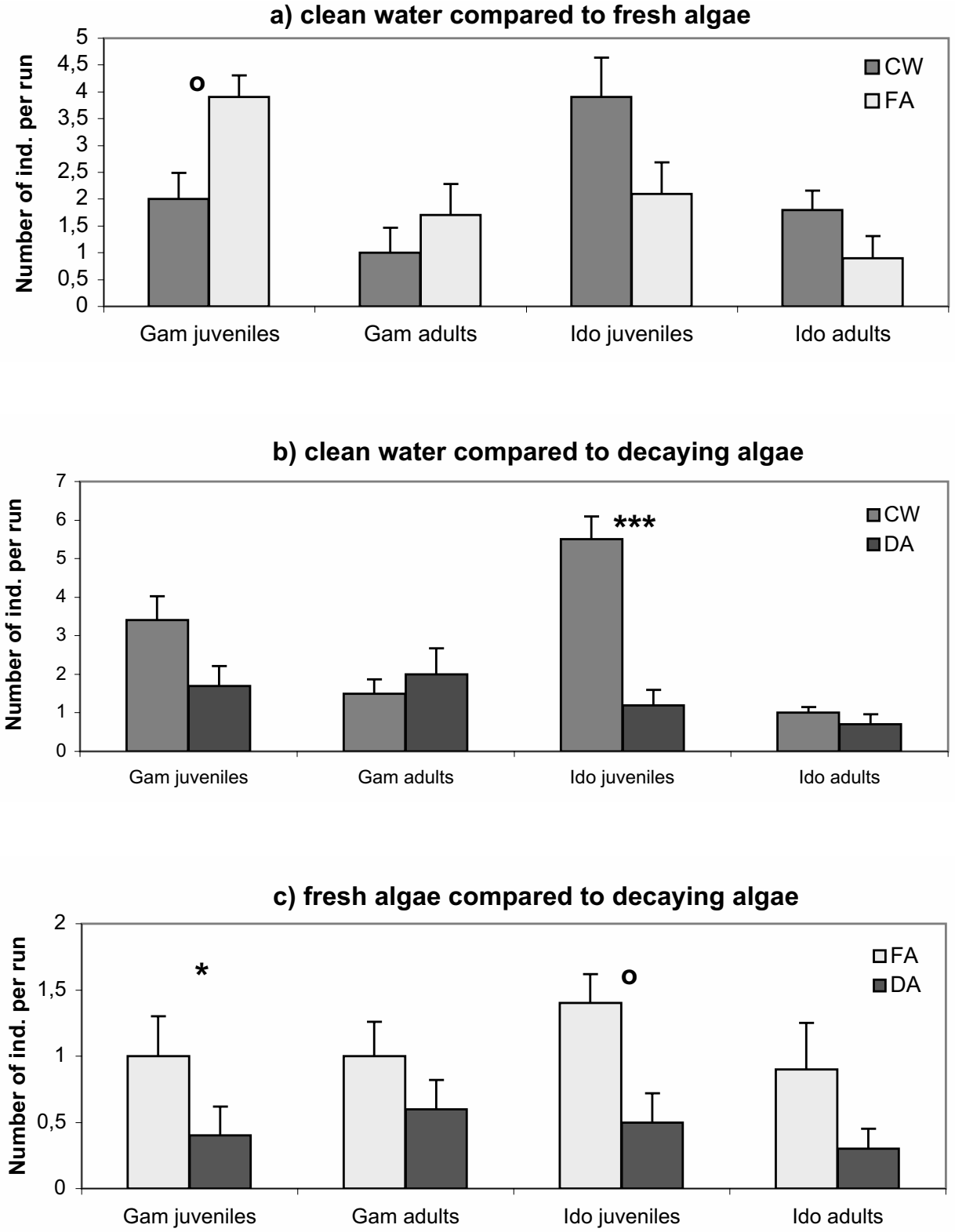

Fig. 4. *Gammarus* spp. and *Idotea baltica*. Preference experiment among juvenile and adult crustacea in Y-aquarium. Means ± SE are given, CW = Clean Sea-Water, FA = water from Fresh Algae and DA = water from Decaying Algae, * means that $0.01 <$ corrected $p \leq 0.05$ and *** means that the corrected $p \leq 0.001$, ° means that only the uncorrected p-value was significant, df = 1, 9.

of snails and isopod crustaceans and lower numbers of chironomids, oligochaetes and gammarids in the attached algae. Regarding biomass, all taxa except *Gammarus* spp., and the chironomids had higher biomasses in the attached algae. The most obvious temporal changes were decreasing numbers of chironomids, oligochaetes, *Gammarus* spp. and *Jaera albifrons* and increasing numbers of molluscs such as *Hydrobia*, *Cerastoderma glaucum* and *Lymnaea* spp. (SIMPER-results not shown for the Time factor).

Aquarium experiments. The nutrient levels of the tested waters differed largely in the aquarium multiple-choice experiments, while the oxygen levels were high throughout (80-90% saturation). Clean seawater (CW) had a total-N of 260 mg l^{-1} and total-P of 22 mg l^{-1}. Fresh algae (FA) had three times higher nitrogen levels (800 mg l^{-1}) and almost ten times higher phosphorus levels (200 mg l^{-1}). Decaying algae (DA) finally, had a low total-N of 23 mg l^{-1}, but an extremely high total-P of 16000 mg l^{-1}.

There were clear differences in water preference between species and between juvenile and adult individuals of the same species (Fig 4). The preference order for both juvenile and adult *Idotea baltica* was CW > FA > DA with significant differences CW > DA and FA > DA for juvenile *I. baltica*. For juvenile *Gammarus* spp., the preference order was FA > CW > DA. Comparisons of FA with CW and further FA with DA were significant at the 0.05-level for juvenile *Gammarus* spp. Adult gammarids responded somewhat differently to the tested waters, choosing FA > DA > CW, but none of the comparisons were significantly different.

DISCUSSION

The results demonstrate that *Cladophora glomerata* in the Northern Baltic Sea support rich animal communities with more than 40 different taxa of macrofauna represented, total abundance ranging between 4600-88100 individuals m^{-2}, and total biomass between 11-490 g wwt m^{-2}. These numbers can be compared to the 25 taxa, 1100-15600 individuals m^{-2} and 13-270 g wwt m^{-2} found in *Fucus vesiculosus* by Kraufvelin & Salovius (unpubl.). However, the total macrofaunal abundance, biomass and animal diversity of *C. glomerata* dropped (by 28% for abundance, 77% for biomass values and 24% for Shannon-Wiener diversity), when the algae detached from the rocky substrate and started to drift and decay. Among the dominating animal populations, bivalves, snails and isopods were more abundant in the attached than in the drifting algae, whereas to a large extent the amphipods were indifferent. We also verified these results for *Idotea baltica* and *Gammarus* spp. in aquarium preference experiments. Regarding temporal differences, we found a clear and rapid succession, with an increase in molluscs and a decrease in arthropods during summer.

The macrofaunal assemblages in our *Cladophora glomerata* samples were not very different from those reported by Jansson (1967) and Fagerholm (1975). Only a few taxa differed markedly from the results presented by Fagerholm (1975). The taxa that were more common in our algal samples were *Hydrobia* spp. (30 times higher abundance in our samples), *Theodoxus fluviatilis* and *Jaera albifrons* (both three times more common in our samples). *Gammarus* spp., on the other hand, were three times more common in the study by Fagerholm (1975). All other dominating taxa differed within a range less than ± 50%.

Occurrence of *Cladophora glomerata*. The *C. glomerata* biomass (range 20 - 130, grand mean 68 g dwt m^{-2}) in our study area was similar to the range reported at other unpolluted or semi-polluted sites of the northern Baltic Sea (Jansson, 1967, 1974; Roos *et al.*, 2004). However, the values were clearly lower than the range of 130-275 g dwt m^{-2} reported for unpolluted sites by Ruokolahti (1988), not to mention his data from sites close to fish farms, which varied from 255 to 460 g dwt m^{-2}. Roos *et al.* (2004) reported a 99% increase in *Cladophora glomerata* biomass at exposed sites and a 38% increase in biomass at sheltered sites of the Åland Islands between 1970 and 2000. In this context, our *C. glomerata* biomass was surprising, since compared to older literature values they gave no support for a huge general increase in algal biomass. One explanation may be that there is an increase in total algal production, but that this increase is not primarily expressed as denser algal belts. A stimulated continuous growth may be masked by increased loss of filaments (through breaking) and a rapid replacement of the lost tissue by new filaments. Barrón *et al.* (2003) and Bokn *et al.* (2003) have expressed similar ideas and have estimated in their experimental rocky littoral mesocosms the amount of algal export at 210 g C m^{-2} $year^{-1}$, which corresponded to 16% of the total net primary production.

The macrofauna assemblages of attached and drifting *Cladophora*. Varying mobility patterns and active preference of available food and habitat explain differences in the macrofauna assemblages between attached and drifting algae. The low abundance of slowly moving or sessile animals like gastropods and bivalves, in the drifting algae probably depends on difficulties for these species to re-enter the loose-lying algae. Gastropods and bivalves among the attached algae may also have originated from the rocky substrate and were scraped off when the algae were sampled. For isopods, the distribution among attached and drifting algae is less dependent on mobility, although the crawling *Jaera albifrons* may have difficulties in entering floating *Cladophora glomerata*. The occurrence of free-swimming *Idotea baltica* seems more to be a consequence of active choice. This can be expressed either as an avoidance of decaying algae / poor water quality or a clear preference for fresh food items / clean water. The observed higher numbers of *I. baltica* in attached algae is supported by a significant preference of water from fresh algae than of water from decaying algae in the aquarium experiments. Similarly, the results for *Gammarus* spp., (slightly higher abundance in drift algae), are supported by the aquarium results for adult gammarids, where no clear water preferences were shown. Whether the adult gammarids mainly are attracted by the nutrients of the drift algae (the high P-concentrations), a different community of epiphytic diatoms / microflora or simply because drifting algae may be more easily palatable than fresh algae, remains to be demonstrated. According to Pöckl (1995), *Gammarus* spp. do not grow (nor survive) very well on a diet consisting of solely *Cladophora* and *Spirogyra*. Nevertheless, gammarids seem to be able to survive and forage almost everywhere, which also is in accordance with other findings (e.g. Fenchel & Kolding, 1979; Christie & Kraufvelin, 2003; Salovius & Bonsdorff, 2004).

The different results for *Idotea baltica* and *Gammarus* spp. may be due to the former species being herbivorous (Kotta *et al.*, 2000; Hemmi & Jormalainen, 2002) with the latter mainly omnivorous (Pöckl, 1995; Cruz-Rivera & Hay, 2000). Differences between adults and juveniles probably depend on juveniles being more sensitive to poor water quality than adults (e.g. Miller *et al.*, 2002), thereby avoiding water from decaying algae.

The high preference for loose-lying *Cladophora glomerata* among gammarids may be a result of the upper littoral being space-limited rather than nutrient-limited. Food is available in excess, but due to high animal densities, the animals must utilise all extra space they can find, thereby also choosing drift algae whenever such is available. Similar ideas have been expressed for rocky shore ecosystems elsewhere (Dayton, 1971; Norderhaug *et al.*, 2002; Christie & Kraufvelin, 2003).

Along with the differences in the macrofauna assemblages between attached and drifting algae, we found differences between times and locations in the data set. In the univariate analyses this either occurred as significant interactions for a number of variables, e.g. total abundance and Shannon-Wiener diversity, or as significant main effects as for total biomass. Multivariate analyses showed additional significant differences with regard to time and could also list the species contributing most to these differences. A temporal succession with an increase in the amount of molluscs and a decrease in the amount of arthropods is in accordance with the lifecycles of most representatives of these phyla. Similar results have been presented for the *Cladophora*-zone for the Askö-area in Sweden by Jansson (1967), and for the Åland Islands by Fagerholm (1975), but the patterns are also familiar from the Baltic *Fucus*-belt (Haage, 1975; Fagerholm, 1978; Kautsky, 1991). Two special traits about the seasonal succession of macrofauna deserve comments. First, very rapid changes occurred for both attached and drifting algae in the abundance NMDS-ordination (Fig. 3a). Each sampling occasion could easily be distinguished in the ordination scheme, despite the short time intervals between samplings (two weeks). Secondly, differences over time were often bigger than differences between attached and drifting algae and always bigger than differences among locations. We think that the rapid seasonal changes in *Cladophora* macrofauna composition will make it difficult to use these macrofauna assemblages for monitoring of the marine environment. Not only will comparisons among sites and regions be risky, comparisons over time (long-term, inter-annual or seasonal) will also be associated with great uncertainty not to mention the establishment of cause-effect relationships to the ongoing eutrophication process.

***Cladophora glomerata*: Wanted, dead or alive?** The epithet "nuisance algae" may be true in

the perspective of humans registering decreased recreational values at the shores, *Fucus vesiculosus* losing in the competition for space and light, fish suffering from deteriorating spawning and foraging grounds and sediment fauna in deeper areas struggling for their survival below dense anoxic algal mats. This study demonstrates, however, that both attached and drifting *Cladophora glomerata* are preferred media for macroinvertebrates and that *C. glomerata* continues to be an important habitat in the littoral after it has detached from its substrate and started to drift and decay. Our results also accentuate the role of drift algae as an alternative habitat for fauna (Norkko *et al.*, 2000; Salovius & Bonsdorff, 2004) and its role in animal transport (Brooks & Bell, 2001). Both attached and drifting *C. glomerata* support a very rich and diverse animal community for Baltic conditions. This community is in turn of the utmost importance at higher trophic levels, i.e. for fish and birds.

MATERIALS AND METHODS

Field studies. We collected our field samples in the outer Archipelago Sea, SW Finland (between 59°55′3-59°55′9 N and 22°20′8-22°21′4 E) where each summer heavy growth of *Cladophora glomerata* occurs on rocky shores from the surface down to a depth of more than 1 m. The area has been documented to suffer from large quantities of drifting algae, which during their decay consume oxygen and decrease ecosystem health in deeper areas (Vahteri *et al.*, 2000). Filamentous algae flourish from spring to autumn, although the species composition changes somewhat during the season with *Pilayella littoralis* (L.) Kjellm. dominating in spring and autumn and *C. glomerata* dominating during the summer months. Since the summer is the most productive period for rocky shore macrofauna, we focused on *C. glomerata.*

We chose four field sites randomly in our study area. At each locality a sparse sublittoral *Fucus vesiculosus* belt occurred below the *Cladophora glomerata* zone at a depth of 0.8-2.0 m. We sampled *C. glomerata* fortnightly four times during the summer of 1999 (24th June, 9th July, 31st July, 16th August), starting from the first occurrence of drifting *C. glomerata* and ending when a clear decrease in growth could be observed. We collected three replicates of both sessile and drifting *C. glomerata* at each sampling occasion and at each locality (from comparable water depths 0.1-0.5 m). We scraped samples of growing clear-green *C. glomerata* from the rock with a round device (diameter 13.5 cm) for sampling *Cladophora* (figure 7.2. p. 91 in Dybern *et al.* 1976). Loose-lying grey-green to brown, non-photosynthetic *C. glomerata* occurred in various amounts near the growing algae (within <5 m), and we sampled this "littoral drift algae" by enclosing a 5 l net bag with a mesh size of 0.5 mm around the algal material. We preserved all samples in 70% alcohol for further analysis in the laboratory. We identified animals >0.5 mm to family or species level, counted and weighed them (wet weight, wwt). After animal extraction, we dried the algae at 60˚C for 24 h to obtain biomass in g dry weight (dwt).

Aquarium experiments. By examining the orientation (swimming) of crustaceans towards a water of known origin, we tested for preference of water quality. The test waters were: 1) clean seawater (CW), 2) water that had been in contact with fresh algae (FA), and 3) water from decaying algae (DA). We utilised Y-formed flow-through aquaria, where we flushed one test water through one arm and another test water through the opposite arm. We tested all waters in paired combinations. Each of the waters flowed at a rate of 15 cm^3 per minute. At the basal part of the Y-aquarium the mixed water left the aquarium. We analysed test waters for nitrogen and phosphorus content according to EPA 353.2-method (U.S. EPA-600 1983), and we measured the oxygen concentrations (all waters were bubbled with oxygen prior to the experiment). We exposed *Gammarus* spp. (*G. oceanicus* Segerstråle and *G. locusta* L.) and *Idotea baltica* Pallas (both taxa and adults and juveniles separately) to different water qualities and recorded their preferences. We placed test organisms (n = 10) below the junction of the arms of the Y-aquarium and registered which water the crustaceans preferred, i.e. the one they swam towards. Three outcomes of the test were possible: one test water, the other test water or no preference (grazers remained in the basal part of the aquaria). We ran the same tests ten times for each animal group, although we used the same specimen only once. Each individual test continued for 6 minutes, a time that we deemed sufficient for selection in previous pilot studies. After 5 runs we switched the waters in order to account for possible laboratory or operational bias, flushed the pipes with clean water and carried out an additional 5 runs.

Table 4. Mean square estimates and appropriate F-ratios for the analysis of the *C. glomerata* macrofauna community using the model: $X_{ijkl} = \mu + L_i + A_j + T_k + LA_{ij} + LT_{ik} + AT_{jk} + LAT_{ijk} + e_{l(ijk)}$, where L = Locality, A = Algae and T = Time.

Source	df	Mean square estimates	F-ratio versus	df for F
$\mathbf{L}_i$	a-1 = 3	$\sigma^2_e + bn\sigma^2_{LT} + bcn\sigma^2_L$	LT	3, 9
$\mathbf{A}_j$	b-1 = 1	$\sigma^2_e + n\sigma^2_{LAT} + an\sigma^2_{AT} + cn\sigma^2_{LA} + acnk^2_A$	No test	–
$\mathbf{T}_k$	c-1 = 3	$\sigma^2_e + bn\sigma^2_{LT} + abn\sigma^2_T$	LT	3, 9
$\mathbf{LA}_{ij}$	$(a\text{-}1)(b\text{-}1) = 3$	$\sigma^2_e + n\sigma^2_{LAT} + cn\sigma^2_{LA}$	LAT	3, 9
$\mathbf{LT}_{ik}$	$(a\text{-}1)(c\text{-}1) = 9$	$\sigma^2_e + bn\sigma^2_{LT}$	Residual	9, 64
$\mathbf{AT}_{jk}$	$(b\text{-}1)(c\text{-}1) = 3$	$\sigma^2_e + n\sigma^2_{LAT} + an\sigma^2_{AT}$	LAT	3, 9
$\mathbf{LAT}_{ijk}$	$(a\text{-}1)(b\text{-}1)(c\text{-}1) = 9$	$\sigma^2_e + n\sigma^2_{LAT}$	Residual	9, 64
$\mathbf{e}_{l(ijk)}$	$abc(n\text{-}1) = 64$	σ^2_e	–	–
Total	$abcn$-1 = 95			

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Statistical analysis. To test for differences in *Cladophora glomerata* biomass for the factors Location = L and Time = T, as well as the interaction L × T, we applied a 2-way factorial ANOVA with three replicates. Both factors were random, orthogonal to each other and had four levels.

For the comparison of macrofauna assemblages between attached and drifting *Cladophora glomerata* we utilised both univariate and multivariate approaches. To test for differences in macrofauna community and population variables for the main factors and interactions among factors, we applied a 3-factor mixed model ANOVA (Table 4) with three replicates and the random factors Location and Time as above were also joined by the fixed factor Algae = A (orthogonal with two levels). We utilised non-parametric multivariate techniques on both species abundance and species biomass. We tested for differences in community structure between attached and drifting algae over time (using pooled replicates from each of the four stations as replicates) in a 2-way crossed analysis of similarity (ANOSIM). We chose to pool the replicates instead of nesting them, since we preferred the extra information available in the 2-way crossed (factorial) ANOSIM above the more restricted amount of information present in a nested ANOSIM. After the hypotheses testing, we used non-metric multidimensional scaling (NMDS) for a graphical overview of between or within group Bray-Curtis similarities and finally we used similarity percentage breakdown (SIMPER) analysis for sorting out the species that contributed most to the observed dissimilarities (Clarke, 1993).

We analysed the aquarium experiments by a randomised-complete-block design for paired comparisons. We chose this paired analysis due to the problem with non-independency of observations in multiple-choice experiments (Peterson & Renaud, 1989; Denton & Chapman, 1991), i.e. that a specific animal individual cannot be present in more than one treatment simultaneously. We considered it valid to arrange the data as a 2-way ANOVA (Sokal & Rohlf, 1995, p. 352-356) since each observation for one treatment may be paired with one for the other treatment. We set up the treatments (the waters) as fixed factors with 2 levels in each comparison, whereas we considered the aquarium run as a blocked random factor with 10 levels (the 10 runs). We tested each pair of waters, CW-FA, CW-DA and FA-DA, separately for each of the four test variables (juvenile and adult *Idotea baltica*, juvenile and adult *Gammarus* spp.).

We ran the univariate statistical analyses on GMAV5 or SPSS 11.0 (Sokal & Rohlf, 1995; Underwood, 1997), whereas we ran all multivariate analyses on PRIMER 5.2.2. (Clarke & Warwick, 1994). Before running the parametric tests, we checked for normality with Kolmogorov Smirnov´s test and homogeneity of variances with Cochran's C-test. To homogenise variances, we used a log-transformation, ln (x + 1). When appropriate, we applied Student-Neuman-Keuls (SNK) tests for *a posteriori* comparisons of the means. For the non-parametric multivariate techniques, we transformed all data by the square root prior to analysis in order to obtain a reasonable balance between dominant and rare animal species. Finally, we used Hochberg's sequential Bonferroni (Hochberg, 1988) to adjust the overall significance levels to the number of tests performed.

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